

Theorem 1.2.1 do not hold on the line  $y = 0$  for the differential equation  $dy/dx = xy^{1/2}$ , and so it is not surprising, as we saw in Example 4 of this section, that there are two solutions defined on a common interval  $(-h, h)$  satisfying  $y(0) = 0$ . On the other hand, the hypotheses of Theorem 1.2.1 do not hold on the line  $y = 1$  for the differential equation  $dy/dx = |y - 1|$ . Nevertheless, it can be proved that the solution of the initial-value problem  $dy/dx = |y - 1|$ ,  $y(0) = 1$ , is unique. Can you guess this solution?

- (ii) You are encouraged to read, think about, work, and then keep in mind Problem 49 in Exercises 1.2.

## 1.2 Exercises

Answers to selected odd-numbered problems begin on page ANS-1.

In Problems 1 and 2,  $y = 1/(1 + c_1 e^{-x})$  is a one-parameter family of solutions of the first-order DE  $y' = y - y^2$ . Find a solution of the first-order IVP consisting of this differential equation and the given initial condition.

1.  $y(0) = -\frac{1}{3}$
2.  $y(-1) = 2$

In Problems 3–6,  $y = 1/(x^2 + c)$  is a one-parameter family of solutions of the first-order DE  $y' + 2xy^2 = 0$ . Find a solution of the first-order IVP consisting of this differential equation and the given initial condition. Give the largest interval  $I$  over which the solution is defined.

3.  $y(2) = \frac{1}{3}$
4.  $y(-2) = \frac{1}{2}$
5.  $y(0) = 1$
6.  $y(\frac{1}{2}) = -4$

In Problems 7–10,  $x = c_1 \cos t + c_2 \sin t$  is a two-parameter family of solutions of the second-order DE  $x'' + x = 0$ . Find a solution of

the second-order IVP consisting of this differential equation and the given initial conditions.

7.  $x(0) = -1, x'(0) = 8$
8.  $x(\pi/2) = 0, x'(\pi/2) = 1$
9.  $x(\pi/6) = \frac{1}{2}, x'(\pi/6) = 0$
10.  $x(\pi/4) = \sqrt{2}, x'(\pi/4) = 2\sqrt{2}$

In Problems 11–14,  $y = c_1e^x + c_2e^{-x}$  is a two-parameter family of solutions of the second-order DE  $y'' - y = 0$ . Find a solution of the second-order IVP consisting of this differential equation and the given initial conditions.

11.  $y(0) = 1, y'(0) = 2$
12.  $y(1) = 0, y'(1) = e$
13.  $y(-1) = 5, y'(-1) = -5$
14.  $y(0) = 0, y'(0) = 0$

In Problems 15 and 16, determine by inspection at least two solutions of the given first-order IVP.

15.  $y' = 3y^{2/3}, y(0) = 0$
16.  $xy' = 2y, y(0) = 0$

In Problems 17–24, determine a region of the  $xy$ -plane for which the given differential equation would have a unique solution whose graph passes through a point  $(x_0, y_0)$  in the region.

17.  $\frac{dy}{dx} = y^{2/3}$
18.  $\frac{dy}{dx} = \sqrt{xy}$
19.  $x \frac{dy}{dx} = y$
20.  $\frac{dy}{dx} - y = x$
21.  $(4 - y^2)y' = x^2$

22.  $(1 + y^3)y' = x^2$   
 23.  $(x^2 + y^2)y' = y^2$   
 24.  $(y - x)y' = y + x$

In Problems 25–28, determine whether Theorem 1.2.1 guarantees that the differential equation  $y' = \sqrt{y^2 - 9}$  possesses a unique solution through the given point.

25.  $(1, 4)$   
 26.  $(5, 3)$   
 27.  $(2, -3)$   
 28.  $(-1, 1)$

29.

- (a) By inspection, find a one-parameter family of solutions of the differential equation  $xy' = y$ . Verify that each member of the family is a solution of the initial-value problem  $xy' = y$ ,  $y(0) = 0$ .  
 (b) Explain part (a) by determining a region  $R$  in the  $xy$ -plane for which the differential equation  $xy' = y$  would have a unique solution through a point  $(x_0, y_0)$  in  $R$ .

- (c) Verify that the piecewise-defined function

$$y = \begin{cases} 0, & x < 0 \\ x, & x \geq 0 \end{cases}$$

satisfies the condition  $y(0) = 0$ . Determine whether this function is also a solution of the initial-value problem in part (a).

30.

- (a) Verify that  $y = \tan(x + c)$  is a one-parameter family of solutions of the differential equation  $y' = 1 + y^2$ .  
 (b) Since  $f(x, y) = 1 + y^2$  and  $\partial f / \partial y = 2y$  are continuous

everywhere, the region  $R$  in Theorem 1.2.1 can be taken to be the entire  $xy$ -plane. Use the family of solutions in part (a) to find an explicit solution of the first-order initial-value problem  $y' = 1 + y^2$ ,  $y(0) = 0$ . Even though  $x_0 = 0$  is in the interval  $(-2, 2)$ , explain why the solution is not defined on this interval.

- (c) Determine the largest interval  $I$  of definition for the solution of the initial-value problem in part (b).

**31.**

- (a) Verify that  $y = -1/(x + c)$  is a one-parameter family of solutions of the differential equation  $y' = y^2$ .
- (b) Since  $f(x, y) = y^2$  and  $\partial f/\partial y = 2y$  are continuous everywhere, the region  $R$  in Theorem 1.2.1 can be taken to be the entire  $xy$ -plane. Find a solution from the family in part (a) that satisfies  $y(0) = 1$ . Find a solution from the family in part (a) that satisfies  $y(0) = -1$ . Determine the largest interval  $I$  of definition for the solution of each initial-value problem.

**32.**

- (a) Find a solution from the family in part (a) of Problem 31 that satisfies  $y' = y^2$ ,  $y(0) = y_0$ , where  $y_0 \neq 0$ . Explain why the largest interval  $I$  of definition for this solution is either  $(-\infty, 1/y_0)$  or  $(1/y_0, \infty)$ .
- (b) Determine the largest interval  $I$  of definition for the solution of the first-order initial-value problem  $y' = y^2$ ,  $y(0) = 0$ .

**33.**

- (a) Verify that  $3x^2 - y^2 = c$  is a one-parameter family of solutions of the differential equation  $ydy/dx = 3x$ .

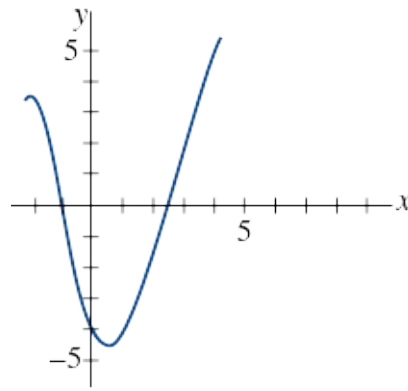
- (b) By hand, sketch the graph of the implicit solution  $3x^2 - y^2 = 3$ . Find all explicit solutions  $y = \phi(x)$  of the DE in part (a) defined by this relation. Give the interval  $I$  of definition of each explicit solution.
- (c) The point  $(-2, 3)$  is on the graph of  $3x^2 - y^2 = 3$ , but which of the explicit solutions in part (b) satisfies  $y(-2) = 3$ ?

**34.**

- (a) Use the family of solutions in part (a) of Problem 33 to find an implicit solution of the initial-value problem  $ydy/dx = 3x$ ,  $y(2) = -4$ . Then, by hand, sketch the graph of the explicit solution of this problem and give its interval  $I$  of definition.
- (b) Are there any explicit solutions of  $ydy/dx = 3x$  that pass through the origin?

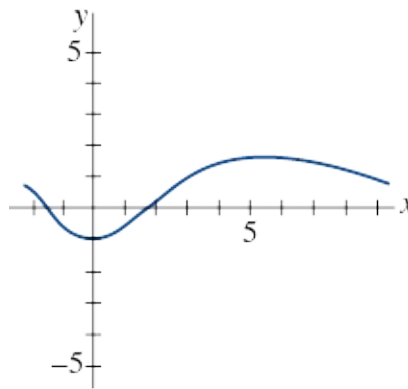
In Problems 35–38, the graph of a member of a family of solutions of a second-order differential equation  $d^2y/dx^2 = f(x, y, y')$  is given. Match the solution curve with at least one pair of the following initial conditions.

- (a)  $y(1) = 1, y'(1) = -2$   
 (b)  $y(-1) = 0, y'(-1) = -4$   
 (c)  $y(1) = 1, y'(1) = 2$   
 (d)  $y(0) = -1, y'(0) = 2$   
 (e)  $y(0) = -1, y'(0) = 0$   
 (f)  $y(0) = -4, y'(0) = -2$



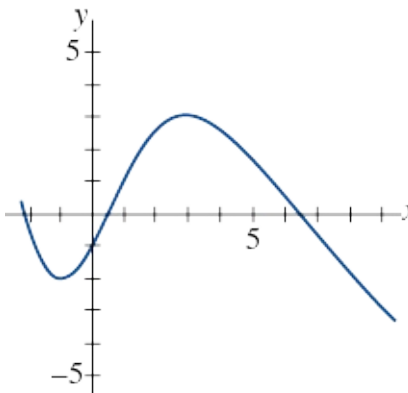
35.

**FIGURE 1.2.7** Graph for Problem 35



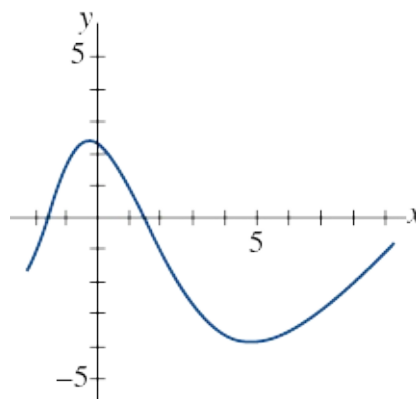
36.

**FIGURE 1.2.8** Graph for Problem 36



37.

**FIGURE 1.2.9** Graph for Problem 37



38.

**FIGURE 1.2.10** Graph for Problem 38

In Problems 39–44,  $y = c_1 \cos 3x + c_2 \sin 3x$  is a two-parameter family of solutions of the second-order DE  $y'' + 9y = 0$ . If possible, find a solution of the differential equation that satisfies the given side conditions. The conditions specified at two different points are called boundary conditions.

39.  $y(0) = 0, y(\pi/6) = -1$

40.  $y(0) = 0, y(\pi) = 0$

41.  $y'(0) = 0, y'(\pi/4) = 0$

42.  $y(0) = 1, y'(\pi) = 5$

43.  $y(0) = 0, y(\pi) = 4$

44.  $y'(\pi/3) = 1, y'(\pi) = 0$

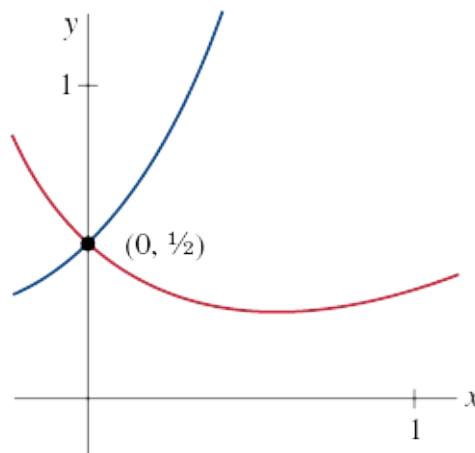
## Discussion Problems

In Problems 45 and 46, use Problem 63 in Exercises 1.1 and (2) and (3) of this section.

45. Find a function  $y = f(x)$  whose graph at each point  $(x, y)$  has the slope given by  $8e^{2x} + 6x$  and has the  $y$ -intercept  $(0, 9)$ .

46. Find a function  $y = f(x)$  whose second derivative is  $y'' = 12x - 2$  at each point  $(x, y)$  on its graph and  $y = -x + 5$  is tangent to the graph at the point corresponding to  $x = 1$ .

47. Consider the initial-value problem  $y' = x - 2y$ ,  $y(0) = \frac{1}{2}$ . Determine which of the two curves shown in **FIGURE 1.2.11** is the only plausible solution curve. Explain your reasoning.



**FIGURE 1.2.11** Graph for Problem 47

48. Without attempting to solve the initial-value problem  $y' = x^2 + y^2$ ,  $y(0) = 1$ , find the values of  $y'(0)$  and  $y''(0)$ .
49. Suppose that the first-order differential equation  $dy/dx = f(x, y)$  possesses a one-parameter family of solutions and that  $f(x, y)$  satisfies the hypotheses of Theorem 1.2.1 in some rectangular region  $R$  of the  $xy$ -plane. Explain why two different solution curves cannot intersect or be tangent to each other at a point  $(x_0, y_0)$  in  $R$ .
50. The functions

$$y(x) = \frac{1}{16}x^4, \quad -\infty < x < \infty$$

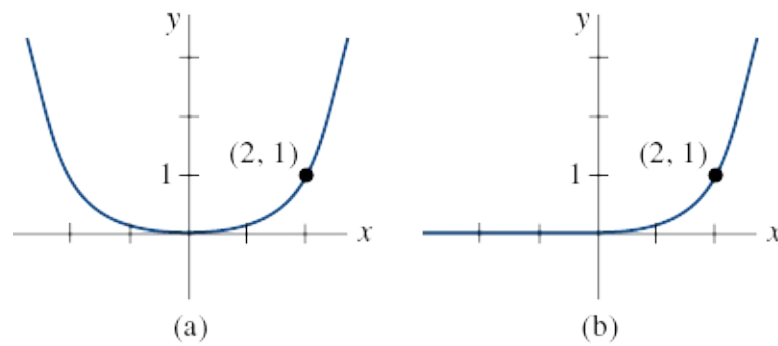
and

$$y(x) = \begin{cases} 0, & x < 0 \\ \frac{1}{16}x^4, & x \geq 0 \end{cases}$$

have the same domain but are clearly different. See **FIGURES 1.2.12(a)** and 1.2.12(b), respectively. Show that both functions are solutions of the initial-value problem  $dy/dx = xy^{1/2}$ ,  $y(2) = 1$  on the interval  $(-\infty, \infty)$ . Resolve the apparent



contradiction between this fact and the last sentence in Example 5.



**FIGURE 1.2.12** Two solutions of the IVP in Problem 50

**51.** Show that

$$x = \int_0^y \frac{1}{\sqrt{t^3 + 1}} dt$$

is an implicit solution of the initial-value problem

$$2 \frac{d^2 y}{dx^2} - 3y^2 = 0, \quad y(0) = 0, y'(0) = 1.$$

Assume that  $y \geq 0$ . [*Hint:* The integral is nonelementary. See (ii) in the *Remarks* at the end of Section 1.1.]